

REINFORCEMENT ANSWERS

SUGGESTED ANSWER FOR CHAPTER 2: MECHANICAL WAVES

NO.	SUGGESTED
1(a)	From the equation $\omega = 30 \text{ rad s}^{-1}$ $A = 0.2 \text{ cm}$ Apply $v_{\max} = \pm \omega A$ $= \pm (30)(0.2)$ $= \pm 6 \text{ cm s}^{-1}$
1(b)	Apply $a_{\max} = A\omega^2$ $= (0.2)(30^2)$ $= 180 \text{ cm s}^{-2}$
2(a)	From Figure 2.1 $A = 8 \text{ cm}$
2(b)	From Figure 2.1 $\frac{3}{4}\lambda = 18 \text{ m}$ $\lambda = 18 \times \frac{4}{3} = 24 \text{ m}$
2(c)	Period, $T = \frac{1}{f} = \frac{1}{20} = 0.05 \text{ s}$
2(d)	Velocity of propagation; $v = f\lambda$ $v = (20)(24)$ $v = 480 \text{ m s}^{-1}$
3	From $T = \frac{n}{t} = \frac{50}{30} = 1.67 \text{ s}$ $f = \frac{1}{T} = \frac{1}{1.67}$ $f = 0.6 \text{ Hz}$

	From $v = \frac{s}{t} = \frac{300}{45} = 6.67 \text{ m s}^{-1}$ From $v = f\lambda$ $\lambda = \frac{v}{f} = \frac{6.67}{0.6} = 11.12 \text{ m}$
4(a)	From $k = \frac{2\pi}{\lambda}$ $\frac{\pi}{6} = \frac{2\pi}{\lambda}$ $\lambda = 12 \text{ cm}$
4(b)	From $\omega = 2\pi f$ $20\pi = 2\pi f$ $f = \frac{20\pi}{2\pi}$ $f = 10 \text{ Hz}$
4(c)	From velocity of wave propagation; $v = f\lambda$ $v = (10)(12)$ $v = 120 \text{ cm s}^{-1}$
5	From wave number; $k = \frac{2\pi}{\lambda}$ $2\pi = \frac{2\pi}{\lambda}$ $\lambda = 1 \text{ m}$ From angular frequency, $\omega = \frac{2\pi}{T}$ $\pi = \frac{2\pi}{T}$ $T = 2 \text{ s}$

5(a)	Graph of $y - x$ at $t = 0$ for $0 \leq x \leq 2\pi$
5(b)	Graph of $y - t$ at $x = 0$ when $0 \leq t \leq 2T$
6(a)	From FIGURE 2.2, Amplitude, $A = 4 \text{ cm}$
6(b)	From FIGURE 2.2, Wavelength, $\lambda = 4 \text{ m}$
6(c)	Wave number, $k = \frac{2\pi}{\lambda}$ $k = \frac{2\pi}{4}$ $k = 0.5 \pi \text{ m}^{-1}$
6(d)	From FIGURE 2.3, Period, $T = 20 \text{ s}$
6(e)	Frequency, $f = \frac{1}{T}$ $f = 0.05 \text{ Hz}$
6(f)	Angular frequency, $\omega = \frac{2\pi}{T}$ $\omega = \frac{2\pi}{20}$ $\omega = \frac{\pi}{10} \text{ rad s}^{-1} = 0.3142 \text{ rad s}^{-1}$

6(g)	Velocity, $v = f\lambda$ $v = (0.05)(4)$ $v = 0.2 \text{ m s}^{-1}$
7	From $y = A \sin(\omega t - kx)$ $A = 10 \text{ cm}; \lambda = 40 \text{ cm}; f = 8 \text{ Hz}$ Wave number, $k = \frac{2\pi}{\lambda} = \frac{2\pi}{40} = 0.05 \pi \text{ cm}^{-1}$ Angular frequency, $\omega = 2\pi f = 2\pi(8) = 16 \pi \text{ rad s}^{-1}$ Substitute A, ω and k into $y = A \sin(\omega t - kx)$ $y = 10 \sin(16 \pi t - 0.05 \pi x)$ where x and y are in cm and t is in s
8(a)	$v = 10 \text{ m s}^{-1}; A = 12 \times 10^{-3} \text{ m}; \lambda = 40 \times 10^{-2} \text{ m}$ $k = \frac{2\pi}{\lambda}$ $= \frac{2\pi}{40 \times 10^{-2}}$ $= 5\pi \text{ m}^{-1}$ Angular frequency; $\omega = 2\pi f = 2\pi \left(\frac{v}{\lambda}\right)$ $= 2\pi \left(\frac{10}{0.4}\right)$ $= 50\pi \text{ rad s}^{-1}$ The equation is $y = (12 \times 10^{-3}) \sin(50\pi t - 5\pi x)$ where y and x are in m and t is in s.
8(b)	Particle vibrational velocity $\frac{dy}{dt} = v = A\omega \cos(\omega t - kx)$ $v = A\omega \cos(\omega t - kx)$ $v = 0.012(50\pi) \cos[(50\pi)(4) - (5\pi)(0.005)]$ $v = 1.196 \times 10^{-2} \text{ m s}^{-1}$

9(a)	From general equation $y = A \sin (\omega t - kx)$ Amplitude, $A = 12 \text{ cm}$ Wavelength, $\lambda = 35 \text{ cm}$ Velocity, $v = f\lambda$ $6 = f (0.35)$ $f = 17.14 \text{ Hz}$ Angular frequency $\omega = 2\pi f$ $= 2\pi(17.14)$ $= 107.71 \text{ rad s}^{-1}$ Wave number, $k = \frac{2\pi}{\lambda}$ $= \frac{2\pi}{0.35}$ $= 17.95 \text{ m}^{-1}$ Substitute the value of A, ω and k into the general equation; $y = 0.12 \sin (107.7t + 17.95x)$ where y and x are in m and t in s
9(b)	Substitute $x = 0.1 \text{ m}$ and $t = 0.5 \text{ s}$ into the wave equation $y = 0.12 \sin (107.7t + 17.95x)$ $y = 0.12 \sin (107.7[0.5] + 17.95[0.1])$ $y = -0.117 \text{ m}$ Substitute $x = 0.1 \text{ m}$ and $t = 0.5 \text{ s}$ into the equation $v = A\omega \cos (\omega t + kx)$ $v = 0.12(107.7)\cos[(107.7)(0.5) + (17.95)(0.1)]$ $v = \pm 2.80 \text{ m s}^{-1}$
9(c)	To the left
10(a)(i)	There are 4 wavelengths in the progressive wave Therefore $4\lambda = 0.2 \text{ m}$ $\lambda = \frac{0.2}{4} = 0.05 \text{ m}$ From $k = \frac{2\pi}{\lambda}$ $= \frac{2\pi}{0.05} = 40 \pi \text{ m}^{-1}$

10(a)(ii)	From $v = f\lambda$ $22 = f(0.05)$ $f = 440 \text{ Hz}$
10(b)	From the general equation $y = A \sin(\omega t - kx)$ Amplitude, $A = \frac{0.05}{2} = 0.025 \text{ m}$ Angular frequency, $\begin{aligned}\omega &= 2\pi f \\ &= 2\pi(440) \\ &= 880\pi \text{ rad s}^{-1}\end{aligned}$ Substitute the A, ω and k into the general equation $y = A \sin(\omega t - kx)$ $y = 0.025 \sin(880\pi t - 40\pi x)$ Where y and x are in m and t is in s.